THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A concave mirror has focal length 15 cm. Its radius of curvature is: 1

(a) 15 cm **(b)** 45 cm **(c)** 7.5 cm **(d)** 30 cm

2. The lens formula is: 1

(a) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ **(b)** $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ **(c)** $v - u = f$ **(d)** $uv = f$

3. The image formed by a plane mirror is: 1

(a) real and inverted **(b)** virtual and erect
(c) real and erect **(d)** virtual and inverted

4. A student states: “A concave mirror can form a virtual image.” The correct response is: 1

(a) Correct **(b)** Only at C
(c) Incorrect **(d)** Cannot be determined

5. The power of a concave lens of focal length 20 cm is: 1

(a) −5 D **(b)** −0.05 D **(c)** −20 D **(d)** +5 D

6. The human eye forms the image of an object at its: 1

(a) pupil **(b)** retina **(c)** cornea **(d)** iris

7. Most of the refraction of light entering the eye occurs at the: 1

(a) retina **(b)** pupil
(c) lens **(d)** outer surface of the cornea

8. The far point of a myopic eye is: 1

(a) less than infinity **(b)** beyond infinity
(c) at 25 cm only **(d)** at infinity

9. A student states: “The colour of the clear sky is blue because blue light has the shortest wavelength of all.” The correct response is: 1

(a) Incorrect — violet is shorter; our eyes are more sensitive to blue
(b) Correct
(c) Correct due to oceans
(d) Cannot be determined

10. A person needs a lens of power −4.5 D. The defect is: 1

(a) myopia **(b)** cataract **(c)** hypermetropia **(d)** presbyopia

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the type of mirror used in 2

(i) headlights

(ii) side mirrors of vehicles.

12. What is the magnification of an image that is twice the size of the object and inverted? 2

13. Name the part of the eye that 2
(i) controls the amount of light
(ii) changes the focal length.

14. Why does a person with cataract have blurred vision? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. An object 2 cm high is placed at 30 cm from a concave mirror of focal length 10 cm. Find the image position and height. 3

16. A person cannot see objects beyond 1.2 m clearly. Name the defect, the type and power of lens needed. 3

17. Find the position of the image when an object is placed 12 cm from a convex lens of focal length 8 cm. Is it real or virtual? 3

OR

State the rules for drawing ray diagrams for spherical mirrors (any three). 3

18. How is the eye able to see both near and distant objects? 3

OR

Why is the colour of the clear sky blue? Explain with the size of particles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Derive the meaning of magnification for a spherical mirror. 5
(ii) An object is placed 6 cm in front of a concave mirror of focal length 12 cm. Find the position and nature of the image.

OR

Case study: A student used a convex lens of focal length 20 cm and moved a candle to find sharp images on a screen. 5
(i) Where should the candle be placed for a same-size image?
(ii) What happens to the image when the candle is moved to 15 cm?
(iii) Find the image distance for the candle at 30 cm.
(iv) Find magnification for (iii).
(v) Power of the lens?

20. (i) Describe the parts of the human eye and their functions. 5
(ii) What is the persistence of vision?

OR

Case study: On a clear day, the sky looks blue but on a foggy day the headlights of cars seem yellow-red. 5
(i) Name the phenomenon.
(ii) Why does the sky appear blue?
(iii) Why are danger signals red?
(iv) Which colour has the longest wavelength?
(v) Why does smoke look bluish in sunlight?

END OF QUESTION PAPER — Total: 40 Marks